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INSTITUTE FOR ECONOMIC DEVELOPMENT



**Reproducible Research Repository**

# Analyzing Data and Generating Reproducible Outputs



# Overview

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## Inputs:

- Constructed datasets:
  - `TZA_CCT_analysis.csv`

## Outputs:

- Raw and Final outputs (Tables/Figures)

## Documentation:

- Collection of reproducible outputs with notes for a README.

## Summary statistics

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**Summary statistics** appear in almost every research product

- Used to highlight central tendencies (means/medians)
- Describe variation in data and comparison groups (SE/SD)
- Indicate ranges and outliers (quantiles and observation numbers)

## Exercise 1

### Exercise 1: Create a summary statistics table

1. Create a summary statistics table with the mean and SD of the variables `hh_size`, `n_child_5`, `food_cons_usd_w`, `area_acre_w`, `crop_damage` by district
2. Export the result in a CSV file or the format of your choice

## Regression tables

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# Regression tables report comparisons

**Regression tables** appear in almost every academic research product

- Used to indicate the exact comparisons being drawn in research
- Describe control variables and outcomes for each model
- Contextualize regression comparisons with more information

### Exercise 2: Create a regression table

1. First, construct a basic regression model to examine whether the treatment has an effect on the value of food consumption.
2. Add controls for other characteristics one at a time. Start by controlling for crop damage and drought/floods.

# Plots

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## Plots provide quick looks

**Plots** can be used to quickly scan data for patterns and problems

- Often use one continuous variable and several categories
- Typically do not develop detailed correlations

## Exercise 3

### Exercise 3: Create a histogram chart

Create a histogram of the `area_crop_w` variable for the treatment group.

## Exercise 4

Exercise 4: Save your plot

Save the resulting plot as png.

# THANK YOU



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